

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 9
9 March – 15 March 2026

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12 January – 8 December 2026



Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



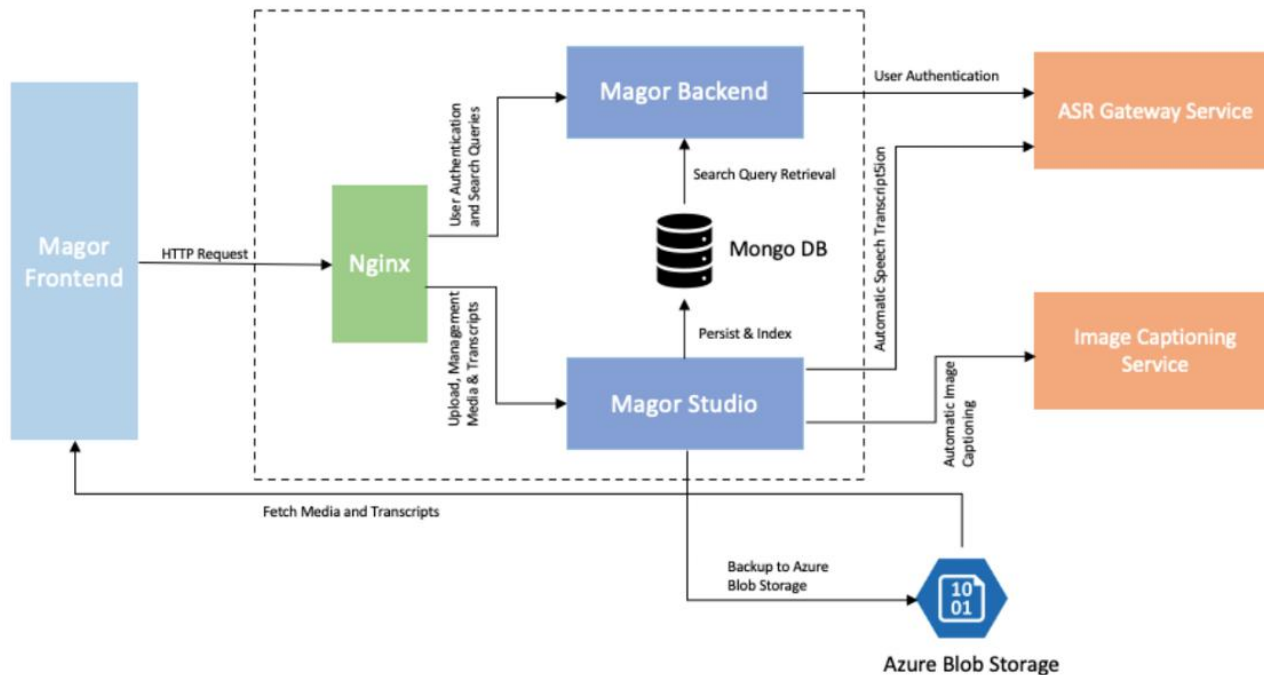
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

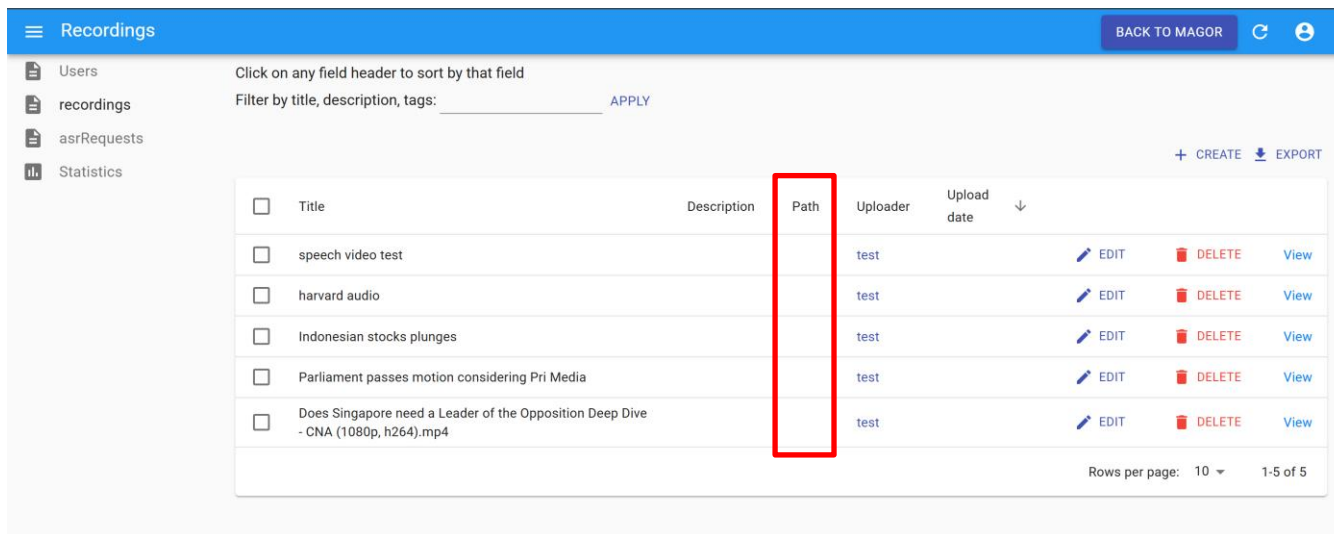
Work done this week:

1. Admin Recording Management Page:
 - Fixed path display and table overflow issue
 - Removed create button
2. Admin User Management Page:
 - Fixed user status bugs
 - Implemented frontend validators



Task 1 (Restore Admin Recording Page Functionality)

- Fixed path display



The screenshot shows the 'Recordings' page in an admin interface. It features a sidebar with navigation links: Users, recordings, asrRequests, and Statistics. The main content area has a filter bar and a table of recordings. The 'Path' column in the table is highlighted with a red box.

☐	Title	Description	Path	Uploader	Upload date	↓
☐	speech video test			test		EDIT DELETE View
☐	harvard audio			test		EDIT DELETE View
☐	Indonesian stocks plunges			test		EDIT DELETE View
☐	Parliament passes motion considering Pri Media			test		EDIT DELETE View
☐	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4			test		EDIT DELETE View

Rows per page: 10 1-5 of 5

Before



Task 1 (Restore Admin Recording Page Functionality)

1. Fixed path display

- Fixed data mapping mismatch

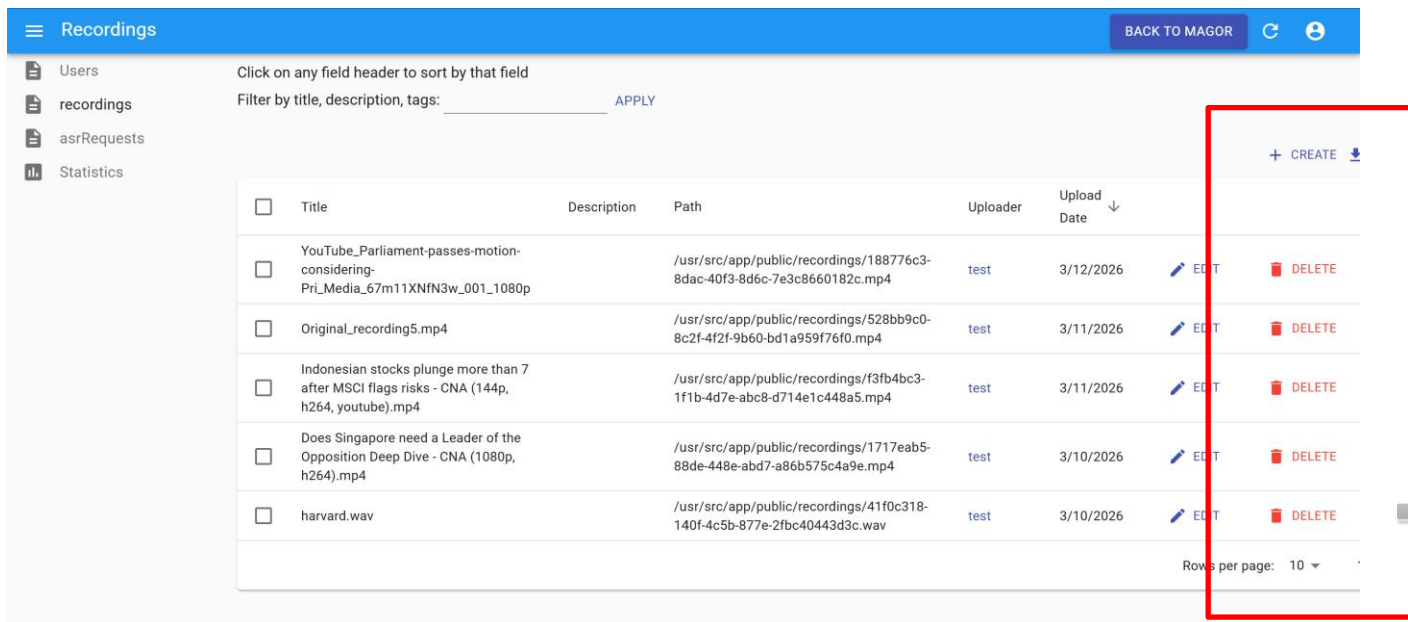
```
<TextField source="path" />
33+
34+ <FunctionField
35+   label="Path"
36+   render={(record) =>
37+     record?.tempResourceUrl ||
38+     record?.azureResourceUrl ||
39+     record?.path ||
40+     record?.url ||
41+     '-'
42+   }
  />
```

Path Data Mapping



Task 1 (Restore Admin Recording Page Functionality)

- Fixed path display
 - Long paths caused table overflow beyond screen width



Recordings

Click on any field header to sort by that field

Filter by title, description, tags: [APPLY](#)

[+ CREATE](#)

☐	Title	Description	Path	Uploader	Upload Date		
☐	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p		/usr/src/app/public/recordings/188776c3-8dac-40f3-8d6c-7e3c8660182c.mp4	test	3/12/2026	EDIT	DELETE
☐	Original_recording5.mp4		/usr/src/app/public/recordings/528bb9c0-8c2f-4f2f-9b60-bd1a959f76f0.mp4	test	3/11/2026	EDIT	DELETE
☐	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4		/usr/src/app/public/recordings/f3fb4bc3-1f1b-4d7e-abc8-d714e1c448a5.mp4	test	3/11/2026	EDIT	DELETE
☐	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4		/usr/src/app/public/recordings/1717eab5-88de-448e-abd7-a86b575c4a9e.mp4	test	3/10/2026	EDIT	DELETE
☐	harvard.wav		/usr/src/app/public/recordings/41f0c318-140f-4c5b-877e-2fbc40443d3c.wav	test	3/10/2026	EDIT	DELETE

Rows per page: 10

Task 1 (Restore Admin Recording Page Functionality)

- Fixed path display
 - Controlled content length for better display

The screenshot shows the 'Recordings' page in an admin interface. The table lists recordings with columns for Title, Description, Path, Uploader, and Upload Date. Two red boxes highlight specific areas: one around the 'Description' column header and the first row's description, and another around the 'Path' column header and the first row's path. A red arrow points from the text 'Description ellipsis' to the first row's description. Another red arrow points from the text 'Only show file name (hover for full path)' to the first row's path. The path is truncated to show only the file name, with a tooltip displaying the full path on hover.

Recordings Table:

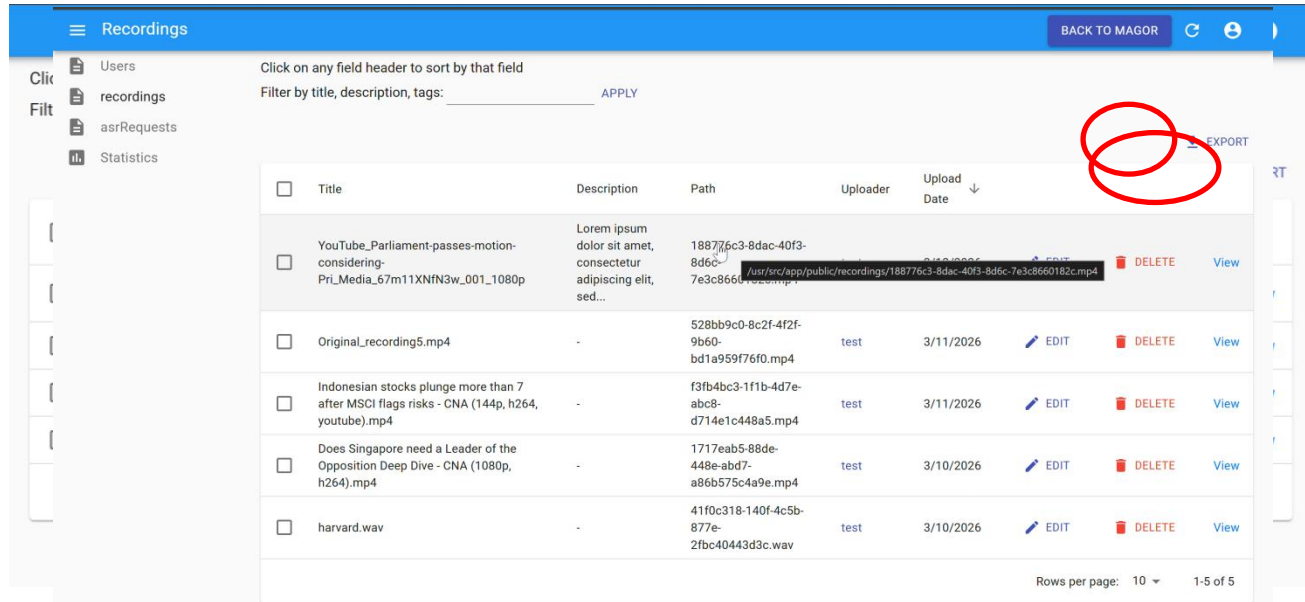
☐	Title	Description	Path	Uploader	Upload Date		
☐	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed...	188776c3-8dac-40f3-8d6c-7e3c8660182c.mp4			DELETE	View
☐	Original_recording5.mp4	-	528bb9c0-8c2f-4f2f-9b60-bd1a959f76f0.mp4	test	3/11/2026	EDIT	DELETE View
☐	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	-	f3fb4bc3-1f1b-4d7e-abc8-d714e1c448a5.mp4	test	3/11/2026	EDIT	DELETE View
☐	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	-	1717eab5-88de-448e-abd7-a86b575c4a9e.mp4	test	3/10/2026	EDIT	DELETE View
☐	harvard.wav	-	41f0c318-140f-4c5b-877e-2fbc40443d3c.wav	test	3/10/2026	EDIT	DELETE View

Rows per page: 10 1-5 of 5



Task 1 (Restore Admin Recording Page Functionality)

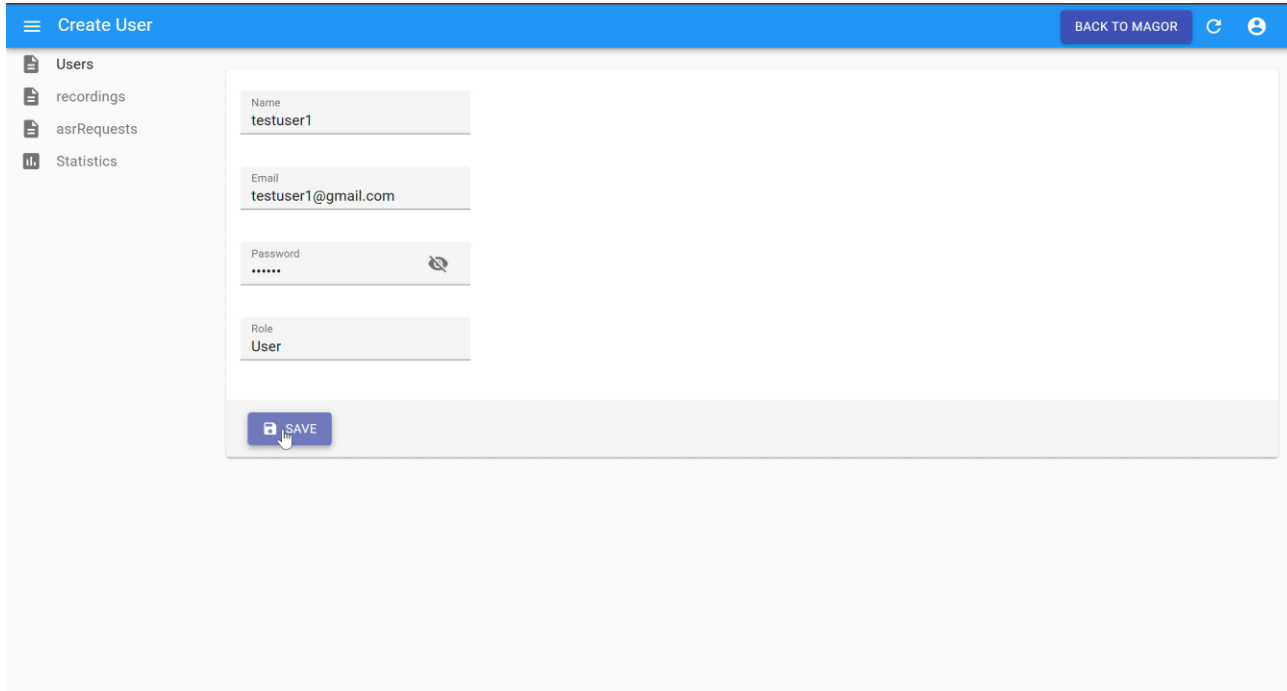
- Removed create button for recordings



With Create Button

Task 2 (Restore Admin User Management Functionality)

- Conflicting success and error responses during user creation



The screenshot shows a web application interface for creating a new user. The top navigation bar is blue and contains a hamburger menu icon, the text 'Create User', a 'BACK TO MAGOR' button, a refresh icon, and a user profile icon. The left sidebar is light gray and lists four menu items: 'Users' (selected), 'recordings', 'asrRequests', and 'Statistics'. The main content area is white and contains a form with the following fields: 'Name' with the value 'testuser1', 'Email' with the value 'testuser1@gmail.com', 'Password' with masked characters and a toggle icon, and 'Role' with the value 'User'. At the bottom of the form is a blue 'SAVE' button with a floppy disk icon. A mouse cursor is hovering over the 'SAVE' button.



Task 2 (Restore Admin User Management Functionality)

- Data mapping mismatch caused error

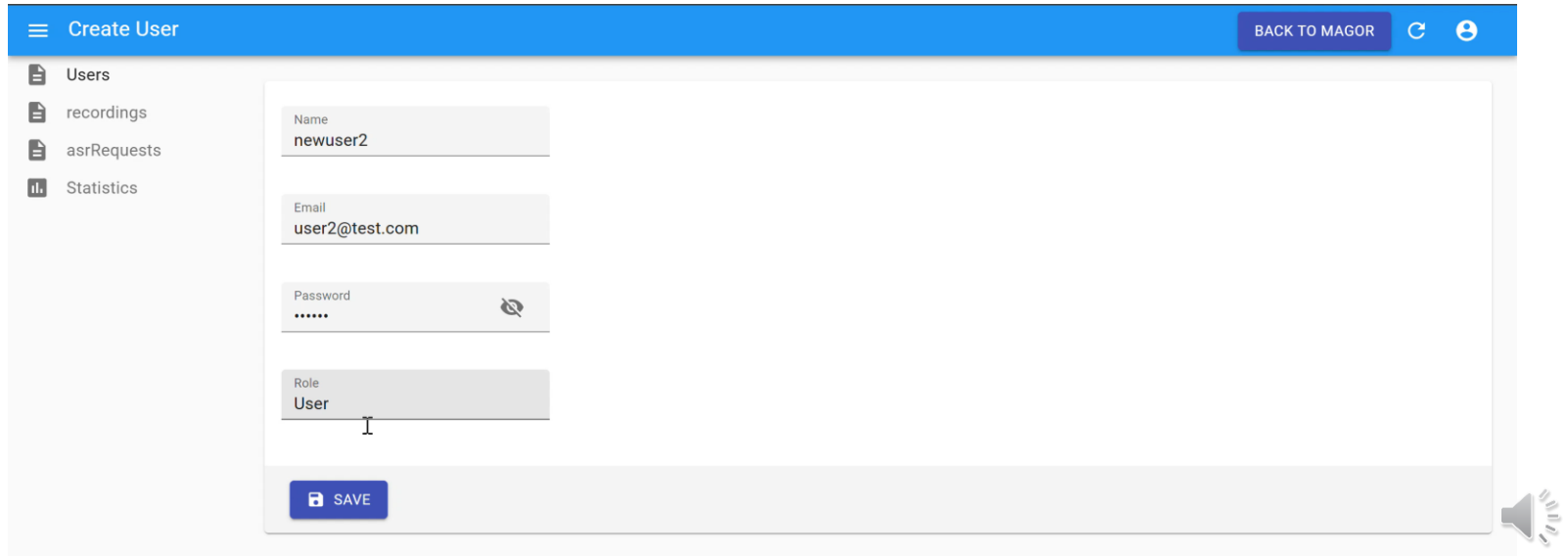
```
create: (resource, params) => {  
  if (resource === 'users') {  
    return httpClient(`${apiUrl}/register`, {  
      method: 'POST',  
      body: JSON.stringify(params.data),  
    }).then(({ json }) => ({  
      data: json,  
      data: { ...json.user, id: json.user._id },  
    })))  
  }  
}
```

Data mapping mismatch



Task 2 (Restore Admin User Management Functionality)

- Fixed user creation demo:



The screenshot shows a web application interface for creating a new user. The top navigation bar is blue and contains a hamburger menu icon, the text 'Create User', a 'BACK TO MAGOR' button, a refresh icon, and a user profile icon. On the left, a sidebar lists navigation options: 'Users' (selected), 'recordings', 'asrRequests', and 'Statistics'. The main content area is a form with four input fields: 'Name' (containing 'newuser2'), 'Email' (containing 'user2@test.com'), 'Password' (masked with dots and having a toggle icon), and 'Role' (containing 'User'). A 'SAVE' button is located at the bottom of the form. A speaker icon is visible in the bottom right corner of the interface.

Create User

BACK TO MAGOR

Users

recordings

asrRequests

Statistics

Name
newuser2

Email
user2@test.com

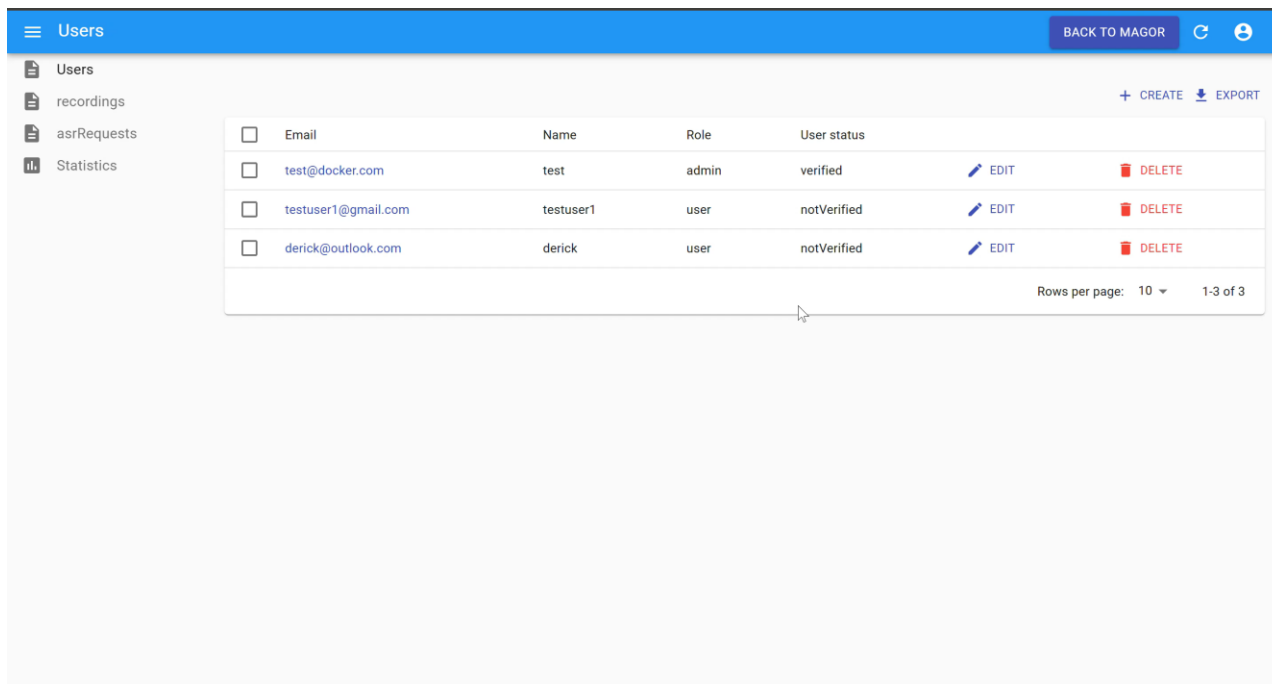
Password
.....

Role
User

SAVE

Task 2 (Restore Admin User Management Functionality)

- Unverified status not displaying correctly



☐	Email	Name	Role	User status		
☐	test@docker.com	test	admin	verified	EDIT	DELETE
☐	testuser1@gmail.com	testuser1	user	notVerified	EDIT	DELETE
☐	derick@outlook.com	derick	user	notVerified	EDIT	DELETE

Rows per page: 10 1-3 of 3



Task 2 (Restore Admin User Management Functionality)

- Unverified status not displaying correctly

```
<AutocompleteInput
  source="userStatus"
  choices={[
    { id: "unverified", name: "Unverified" },
    { id: "notVerified", name: "Unverified" },
    { id: "verified", name: "Verified" },
    { id: "disabled", name: "Disabled" },
  ]}
/>
```

Data mapping mismatch



Task 2 (Restore Admin User Management Functionality)

- Unverified status fixed demo:

Users

recordings

asrRequests

Statistics

BACK TO MAGOR

+

CREATE

↓

EXPORT

☐	Email	Name	Role	User status		
☐	test@docker.com	test	admin	verified	EDIT	DELETE
☐	testuser1@gmail.com	testuser1	user	unverified	EDIT	DELETE
☐	derick@outlook.com	derick	user	unverified	EDIT	DELETE
☐	newuser@test.com	newUser	user	unverified	EDIT	DELETE
☐	new@test.com	newuser	admin	unverified	EDIT	DELETE
☐	user2@test.com	newuser2	user	unverified	EDIT	DELETE
☐	tee@te.com	tee	user	unverified	EDIT	DELETE
☐	derick@hotmail.com	derick	user	unverified	EDIT	DELETE

Rows per page: 10 1-8 of 8



Task 2 (Restore Admin User Management Functionality)

- Missing email format validation

The screenshot shows a web application interface for creating a user. The top navigation bar is blue with the text 'Create User' and a 'BACK TO MAGOR' button. A sidebar on the left lists navigation items: 'Users', 'recordings', 'asrRequests', and 'Statistics'. The main form area contains four input fields: 'Name' with the value 'derick', 'Email' with the value 'derick@dd', 'Password' with masked characters '*****' and a toggle icon, and 'Role' with the value 'User'. A blue 'SAVE' button with a hand icon is positioned below the form fields. At the bottom center of the page, a red error message states: 'Failed to create user. Please try again.'



Task 2 (Restore Admin User Management Functionality)

- Missing password format validation

The screenshot shows a web application interface for creating a user. The top navigation bar is blue with a hamburger menu icon, the text 'Create User', and a 'BACK TO MAGOR' button with a refresh and user icon. The left sidebar contains links for 'Users', 'recordings', 'asrRequests', and 'Statistics'. The main form area has four input fields: 'Name' with the value 'derick', 'Email' with the value 'derick@gmail.com', 'Password' with masked characters '....' and a toggle icon, and 'Role' with the value 'User'. A blue 'SAVE' button is located below the form fields. At the bottom of the page, a red error message box displays the text 'Failed to create user. Please try again.'



Task 2 (Restore Admin User Management Functionality)

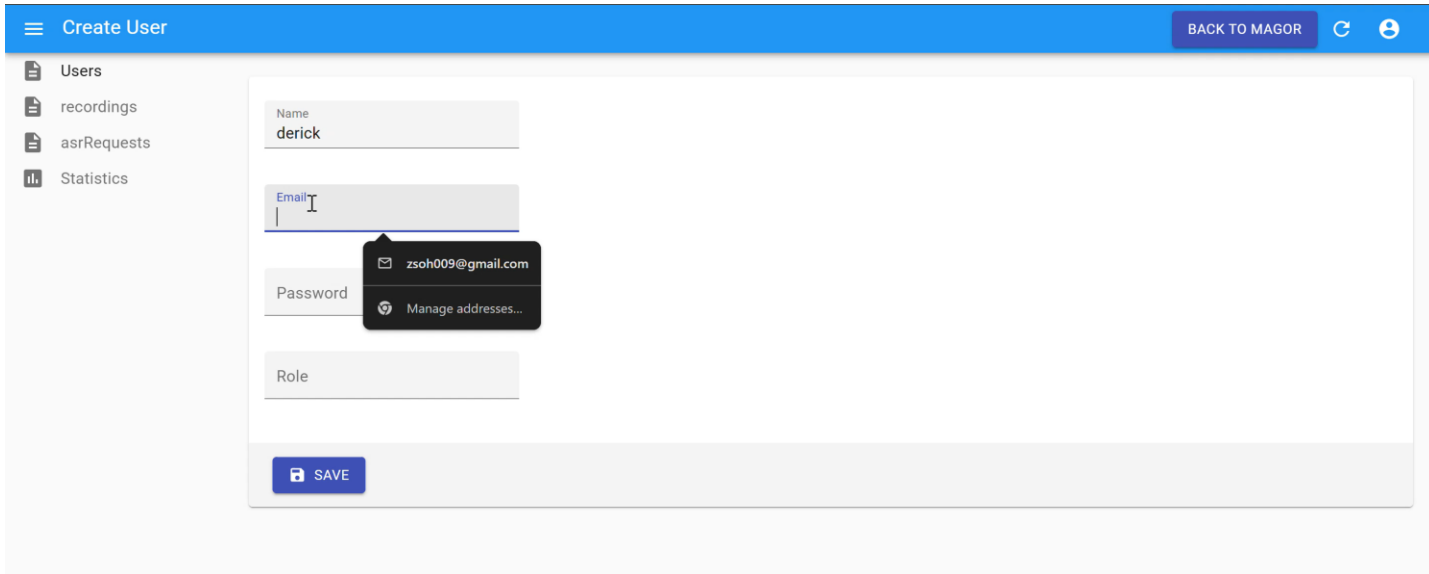
- Added validation functions to form inputs

```
return (  
  <Create title={`Create User`} onSuccess={onSuccess} onFailure={onFailure}  
    <SimpleForm validateOnBlur>  
      <TextInput source="name" />  
      <TextInput source="email" validate={validateEmail} />  
      <PasswordInput source="password" validate={validatePassword} />  
      <AutocompleteInput  
        source="role"  
        choices={[  
          { id: "admin", name: "Admin" },  
          { id: "user", name: "User" },  
          { id: "uploader", name: "Uploader" },  
        ]}  
      />  
    </SimpleForm>  
  </Create>  
)
```



Task 2 (Restore Admin User Management Functionality)

- Input validation demo:



The screenshot displays a web application interface for creating a new user. The top navigation bar is blue and contains a hamburger menu icon, the text 'Create User', a 'BACK TO MAGOR' button, a refresh icon, and a user profile icon. The left sidebar is light gray and lists four menu items: 'Users' (with a document icon), 'recordings' (with a document icon), 'asrRequests' (with a document icon), and 'Statistics' (with a bar chart icon). The main content area is white and contains a form with four input fields: 'Name' (containing 'derick'), 'Email' (containing 'zsoh009@gmail.com'), 'Password', and 'Role'. A dark gray tooltip is visible over the 'Email' field, showing the email address 'zsoh009@gmail.com' and a 'Manage addresses...' link. At the bottom of the form is a blue 'SAVE' button with a floppy disk icon.



Plans for Next Week

- Restore admin page functionality
 - Recording edit page
 - Retrieve and enable transcript edits for recordings
- Migrate file storage from Docker volumes to local paths
 - Verify and update features to align with the new file structure
- Transcript Studio
 - Implement transcript name display and editing

